

Experimental Materials Characterization

Structural, Chemical, and Optical Characterization of
Carbon Fibers with Carbon Nanotubes

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Abstract

This report presents the structural, chemical, and optical characterization of carbon fiber mats reinforced with carbon nanotubes (CNTs), synthesized from a polyacrylonitrile (PAN) precursor via electrospinning and subsequent pyrolysis and chemical vapor deposition (CVD) using iron nanoparticles as catalyst. Multiple characterization techniques were applied to analyze the morphology, chemical composition, crystalline structure, and optical properties of the resulting material.

UV–Vis spectroscopy was used to study the optical properties of the iron oxide (Fe_2O_3) nanoparticles used as catalyst, yielding a band gap estimate of 2.16 eV — consistent with literature values for the hematite phase. X-ray diffraction (XRD) confirmed the crystalline phase of the nanoparticles and enabled estimation of crystallite sizes of approximately 19 nm via the Scherrer equation. FTIR spectroscopy confirmed the successful transformation of the PAN precursor into a carbon-dominated structure through pyrolysis. Raman spectroscopy revealed the characteristic D and G bands of carbonaceous materials, indicative of partially graphitized carbon with a graphitic domain size $L_a \approx 5.02$ nm. Finally, scanning electron microscopy (SEM) and EDS elemental mapping confirmed the formation of CNT-like nanostructures on the fiber surfaces, with a composition dominated by carbon (~ 80 wt%), consistent with the carbonization of PAN and the catalytic action of the iron nanoparticles.

Characterization Techniques

To fully understand the properties of the synthesized material, multiple complementary characterization techniques were employed. The characterization strategy proceeded from the most fundamental component — the iron oxide nanoparticles — toward the complete fiber mat. Techniques applied to the nanoparticles include UV–Vis spectroscopy and X-ray diffraction (XRD). Techniques applied to the carbon fiber mat include FTIR spectroscopy, Raman spectroscopy, and scanning electron microscopy (SEM). These methods are not interchangeable; each provides distinct and complementary information about different aspects of the material.

Iron Oxide Nanoparticle Characterization (Fe_2O_3)

1.1 UV–Vis Spectroscopy

UV–Vis spectroscopy is widely used to study the optical properties of materials. It characterizes how a sample absorbs electromagnetic radiation in the ultraviolet (200–400 nm) and visible (400–800 nm) ranges, providing information about electronic transitions between energy bands.

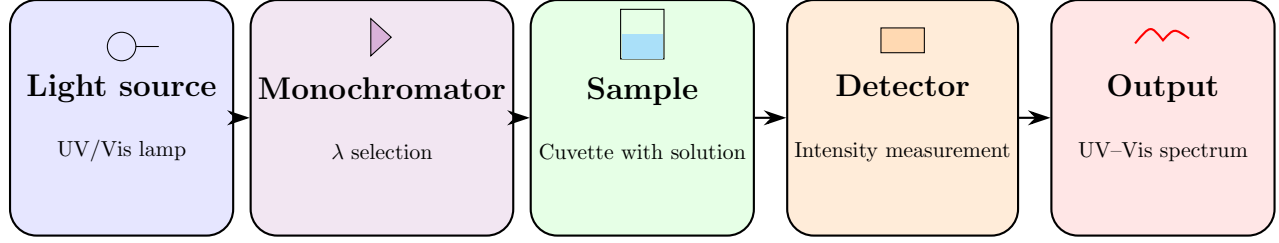


Figure 1: Schematic of a UV-Vis spectrophotometer.

Light absorption is governed by the Beer-Lambert law,

$$A = \log_{10} \left(\frac{I_0}{I} \right) = \varepsilon cl, \quad (1)$$

where A is absorbance, I_0 and I are the incident and transmitted intensities, ε is the molar absorption coefficient, c is the concentration of the absorbing species, and l is the optical path length.

Two measurement modes were used for Fe_2O_3 nanoparticles: transmission (colloidal suspension in Milli-Q water) and diffuse reflectance (solid powder with an integrating sphere attachment). The instrument was an Agilent Cary 7000 Universal Measurement Spectrophotometer.

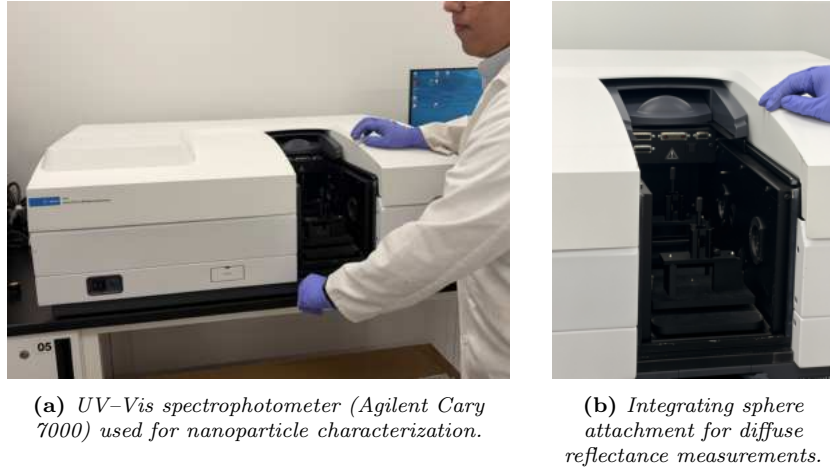


Figure 2: UV-Vis equipment used for optical characterization of Fe_2O_3 nanoparticles.

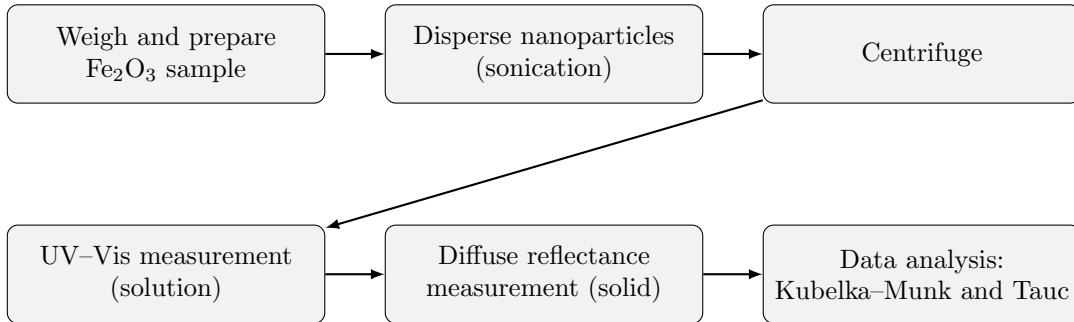


Figure 3: Experimental workflow for UV-Vis characterization of Fe_2O_3 nanoparticles.

Sample preparation procedure. A colloidal solution of Fe_2O_3 at 0.005 mol L^{-1} in 5 mL of Milli-Q water was prepared, requiring approximately 4 mg of nanoparticles (molar mass 159.7 g mol^{-1}). After weighing on an Ohaus Pioneer analytical balance, the nanoparticles were dispersed by sonication (Omni-Ruptor 4000, 1 min at 30% power), then centrifuged at 3600 rpm for 3 minutes (Thermo Scientific ST 40R) to remove larger aggregates. The supernatant was collected for spectroscopic analysis.



Figure 4: *Ohaus Pioneer analytical balance.*



Figure 5: *Omni-Ruptor 4000 sonicator.*



Figure 6: *Thermo Scientific ST 40R centrifuge.*

The diffuse reflectance spectra were recorded as percent reflectance (%R) versus wavelength and converted to the Kubelka–Munk function for band gap estimation:

$$F(R) = \frac{(1 - R)^2}{2R}, \quad (2)$$

where R is the diffuse reflectance. The Tauc method was then applied,

$$(F(R) h\nu)^n = A (h\nu - E_g), \quad (3)$$

where $h\nu$ is the photon energy, E_g is the band gap, A is a constant, and n depends on the type of electronic transition.

1.2 X-ray Diffraction (XRD)

XRD analysis was carried out on Fe_2O_3 nanoparticle powder deposited on a single-crystal silicon substrate, using a Bruker D2 PHASER diffractometer with $\text{Cu}_{K\alpha}$ radiation ($\lambda = 1.54 \text{ \AA}$) in Bragg–Brentano (θ – 2θ) configuration.

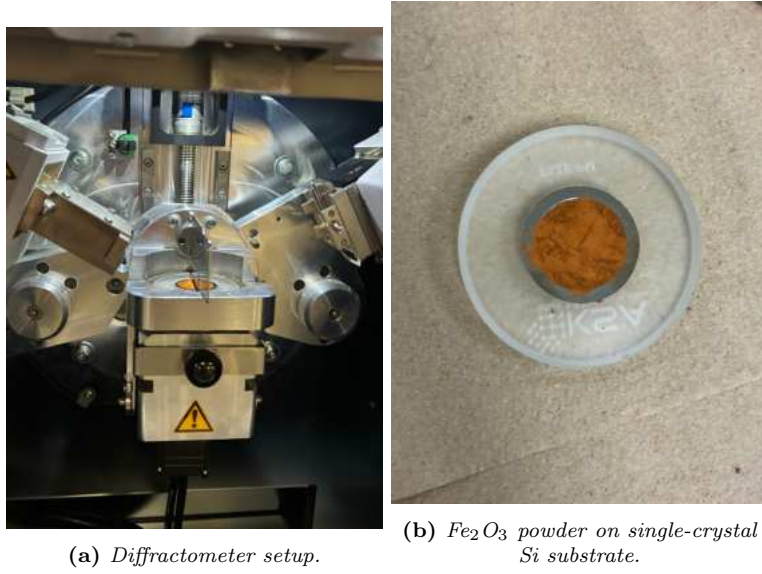


Figure 7: XRD sample preparation and experimental setup.

Two datasets were acquired: a high-resolution pattern ($\Delta(2\theta) = 0.01^\circ$, 1.81 s/step, 5956 steps, 10800 s total) and an experimental pattern ($\Delta(2\theta) = 0.05^\circ$, 1 s/step, 1193 steps, 1291 s total), both in the range 20° – 80° . Soller slits, an anti-scatter slit, a divergence slit, and a Ni β -filter were used to collimate the beam and remove $K\beta$ radiation.

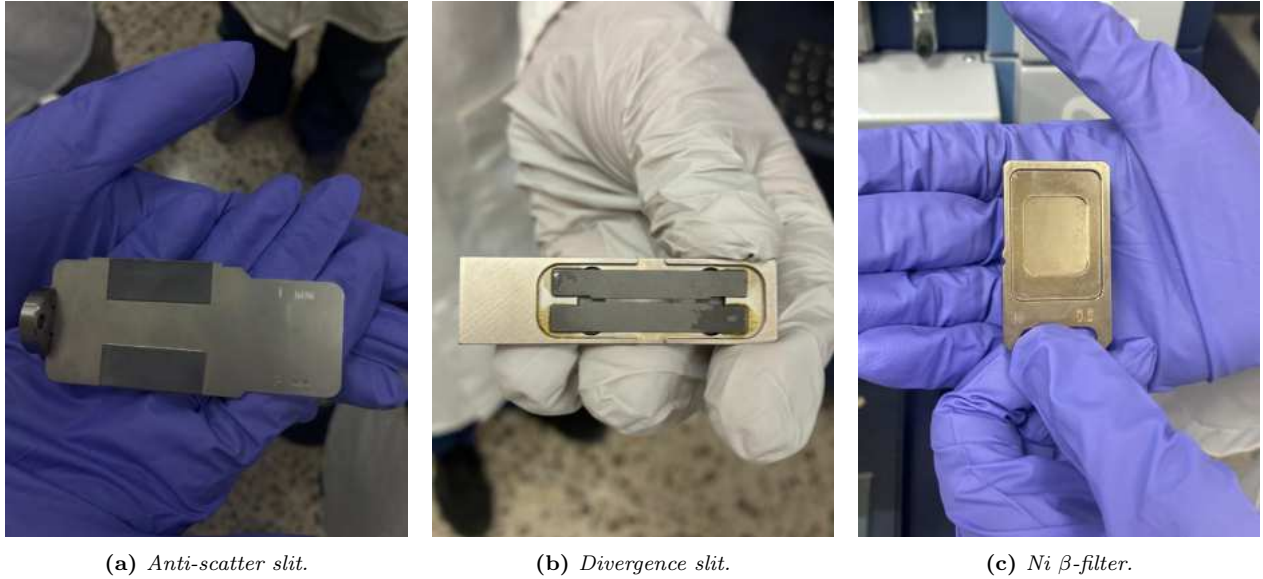


Figure 8: Optical components used in the XRD system.

Three analysis methods were applied to extract structural parameters.

Scherrer equation. Crystallite size D is estimated from peak broadening:

$$D = \frac{K\lambda}{B_L \cos \theta}, \quad (4)$$

where $K = 0.89$, $\lambda = 1.54 \text{ \AA}$, B_L is the FWHM in radians, and θ is the Bragg angle.

Harris method. The texture coefficient quantifies preferential crystallographic orientation by comparing measured intensities to a random reference pattern:

$$T_{c(hkl)} = \frac{I_{(hkl)}/I_{0(hkl)}}{\frac{1}{n} \sum I_{i(hkl)}/I_{i0(hkl)}}, \quad (5)$$

where $T_{c(hkl)} = 1$ corresponds to a randomly oriented sample; values above 1 indicate preferential orientation.

Williamson–Hall method. Total peak broadening B_t is decomposed into size and strain contributions:

$$B_t = B_s + B_L, \quad B_s = 4\varepsilon \tan \theta, \quad (6)$$

yielding the linearized Williamson–Hall equation

$$B_t \cos \theta = \frac{K\lambda}{D} + 4\varepsilon \sin \theta, \quad (7)$$

from which both D and the microstrain ε are extracted by linear regression.

Carbon Fiber Mat Characterization

2.1 FTIR Spectroscopy

Fourier-transform infrared (FTIR) spectroscopy characterizes molecular vibrations through the interaction of infrared radiation with chemical bonds. When the radiation frequency matches a vibrational mode of the material, energy is absorbed. The resulting spectrum identifies functional groups and allows tracking of chemical transformations during thermal processing.

The principal vibrational modes observable by FTIR include:

Vibrational modes	
Stretching	Scissoring
Symmetric stretching	Rocking
Asymmetric stretching	Wagging
Bending	Twisting

Table 1: *Principal vibrational modes observable in FTIR spectroscopy.*

Two samples were analyzed: the PAN precursor and the carbonized mat after CVD and pyrolysis. Spectra were recorded from 400 to 4000 cm^{-1} at 4 cm^{-1} resolution using a PerkinElmer Frontier spectrometer

equipped with a diamond attenuated total reflectance (ATR) accessory. The instrument was cleaned with acetone between measurements.



Figure 9: *PerkinElmer Frontier FTIR spectrometer.*

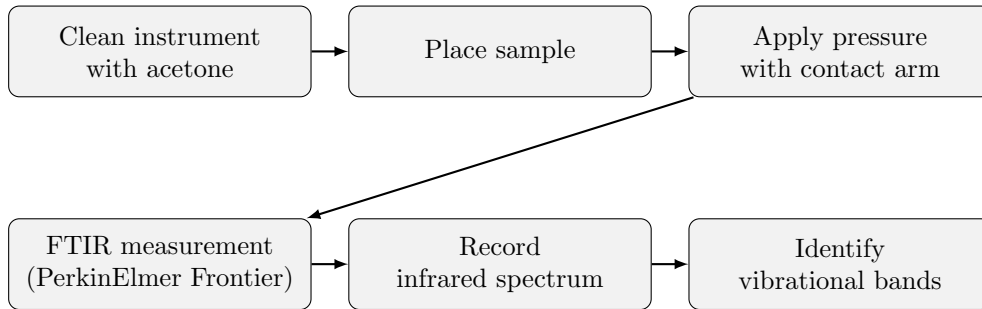


Figure 10: *Experimental workflow for FTIR characterization of the fiber mat samples.*

2.2 Raman Spectroscopy

Raman spectroscopy probes molecular vibrations through inelastic scattering of monochromatic light. Because the Raman scattering cross-section is very small, a high-intensity monochromatic source is required; an argon-ion laser at 514 nm was used. The laser was directed through an optical system onto the sample via a microscope objective; the Rayleigh-scattered component was rejected by notch filters, and only the Raman-scattered photons reached the CCD detector.

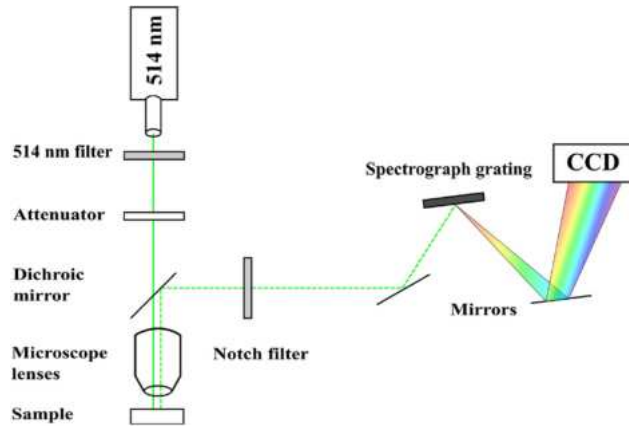


Figure 11: Basic schematic of a Raman spectrometer.

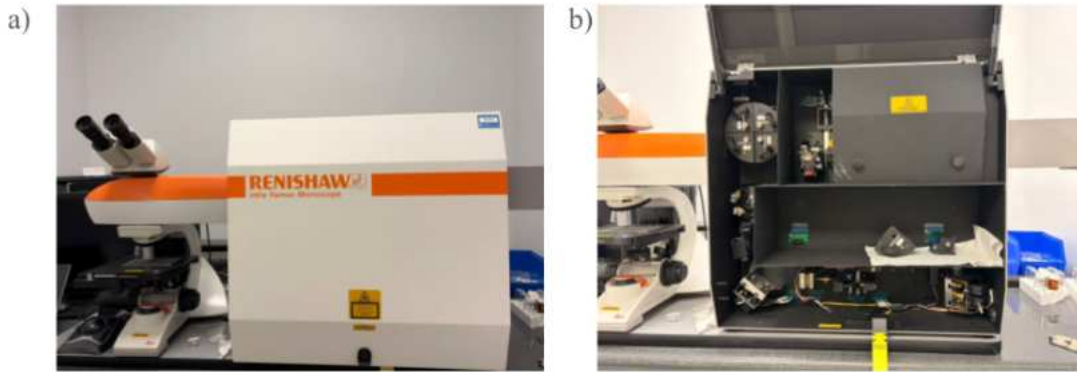


Figure 12: (a) Renishaw Raman spectrometer used. (b) Interior of the optical module.

A point analysis was performed on the carbonized mat in the range $100\text{--}3200\text{ cm}^{-1}$, with 10 s integration time, 3 accumulations, 5% laser power, and a $50\times$ objective. The aluminum sample holder was chosen because it produces no appreciable Raman signal in the measurement range. The laboratory was kept dark and electronic devices were disabled to avoid optical interference.

2.3 Scanning Electron Microscopy (SEM)

SEM was performed on the carbon fibers before and after thermal treatment to assess morphological changes and CNT growth, and to obtain elemental composition via EDS. The instrument was a ZEISS GeminiSEM 560, equipped with a ZrO_2 -coated tungsten field-emission gun operating at $1700\text{--}1800\text{ K}$.



Figure 13: ZEISS GeminiSEM 560.

The instrument operates inside a Faraday cage with anti-vibration platforms and humidity control. Three vacuum pumps (turbomolecular, thermionic, and mechanical) maintain pressures of 1×10^{-10} mbar at the filament and 2×10^{-7} mbar in the chamber. Nitrogen was used to neutralize surface charging; additionally, the fiber samples were sputter-coated with gold to prevent charge accumulation. Secondary and backscattered electron detectors were used at an accelerating voltage of 4 kV, selected to maximize surface topographic contrast while minimizing beam damage.

Results and Discussion

UV–Vis Spectroscopy

Colloidal suspension

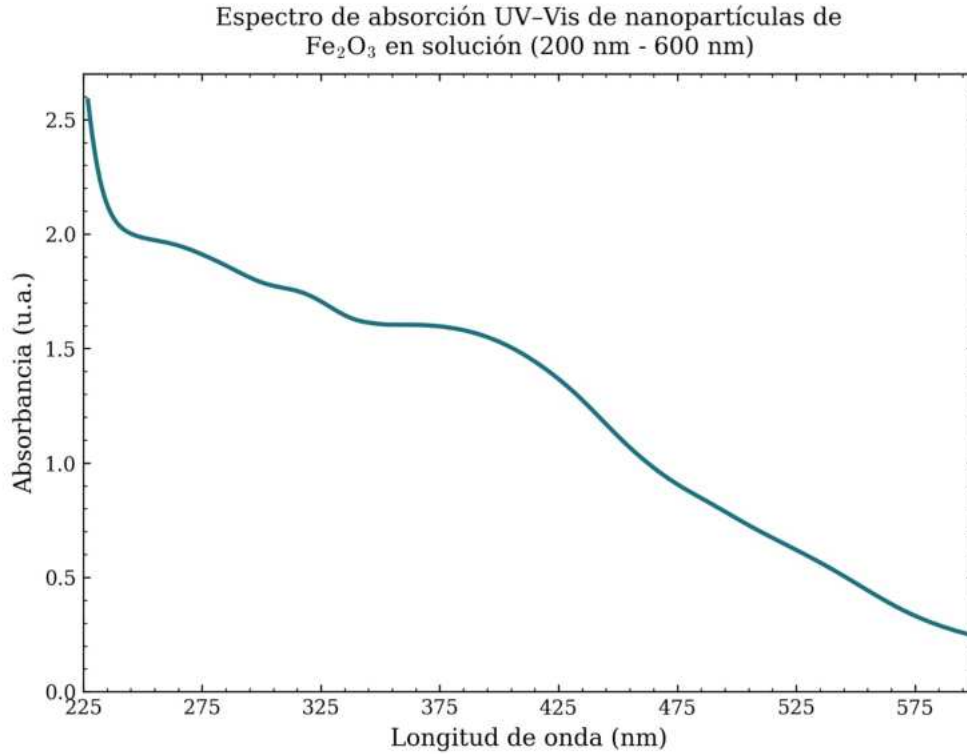
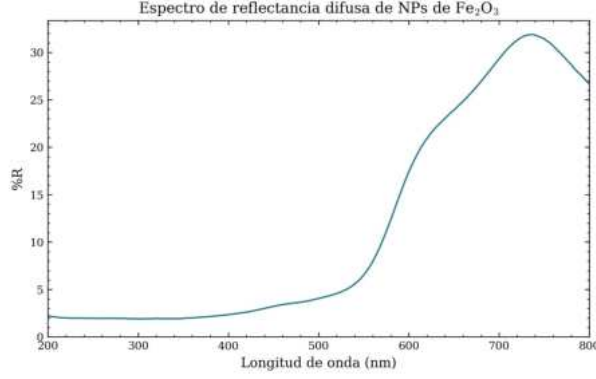


Figure 14: *UV–Vis absorption spectrum of Fe_2O_3 nanoparticles in colloidal suspension (200–600 nm).*

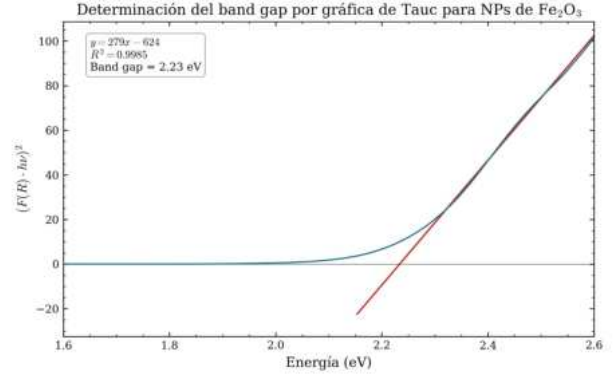
The absorption spectrum (Figure 14) shows a monotonic decay from the UV toward the visible region, characteristic of Fe_2O_3 nanoparticles. This behavior is associated with electronic transitions between the valence and conduction bands, confirming the semiconducting nature of the material. The absorption edge extending into the visible range is directly related to the band gap energy; its exact value was determined by diffuse reflectance measurements.

Solid powder — diffuse reflectance and Tauc analysis

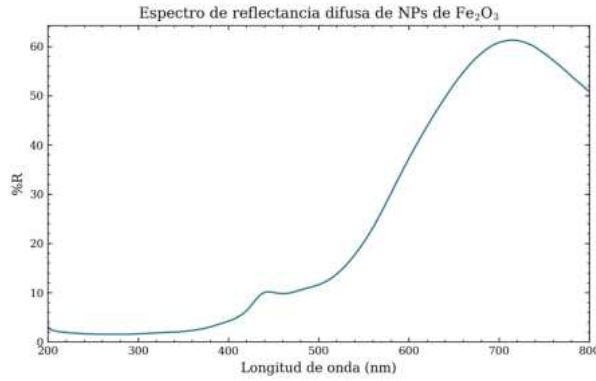
Powder measurements required multiple samplings to obtain adequate signal quality.



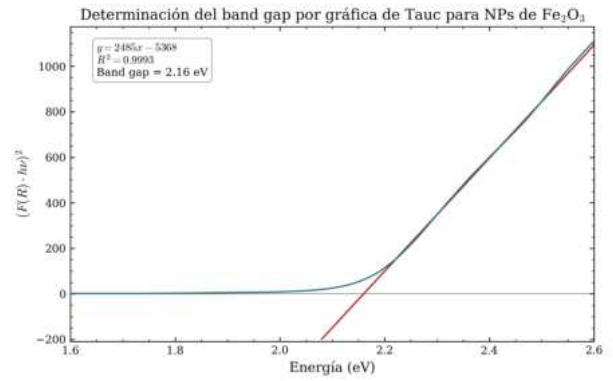
(a) Diffuse reflectance spectrum — first measurement.



(b) Tauc plot — first measurement.

Figure 15: Optical characterization of Fe_2O_3 nanoparticles — first sampling.


(a) Diffuse reflectance spectrum — second measurement.



(b) Tauc plot — second measurement.

Figure 16: Optical characterization of Fe_2O_3 nanoparticles — second sampling.

Both reflectance spectra show increasing reflectance toward the visible region, consistent with iron oxide semiconductors. The Tauc plots were restricted to the spectral region where the absorption onset occurs, and the linear extrapolation to zero yields the band gap estimate.

Sample	Measured E_g (eV)	Literature E_g (eV)	Difference (eV)
Sampling 1	2.23	2.11	0.12
Sampling 2	2.16	2.11	0.05

Table 2: Band gap values obtained by the Tauc method compared to literature values for Fe_2O_3 nanoparticles.

Both measurements fall within the 2.00–2.30 eV range consistent with hematite-phase Fe_2O_3 . Minor differences between samplings are attributed to particle size variations and structural defects within the powder. The second sampling ($E_g = 2.16$ eV) shows better agreement with the literature reference value of 2.11 eV.

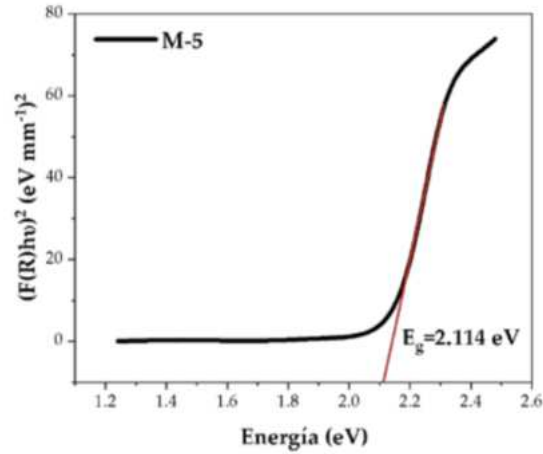


Figure 17: *UV-Vis absorption spectrum reported in the literature for Fe_2O_3 nanoparticles, used as a reference for comparison.*

X-ray Diffraction (XRD)

Phase identification

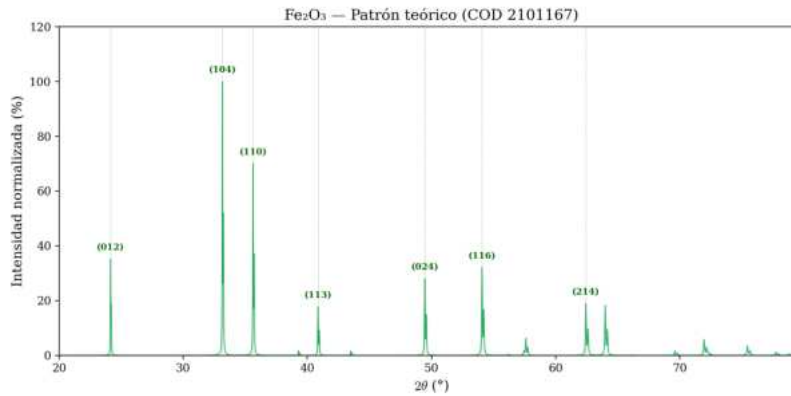
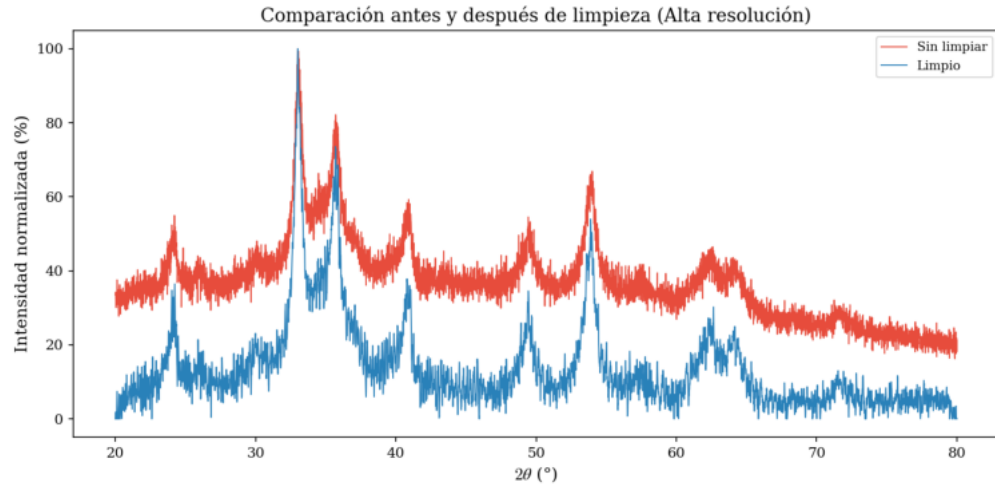
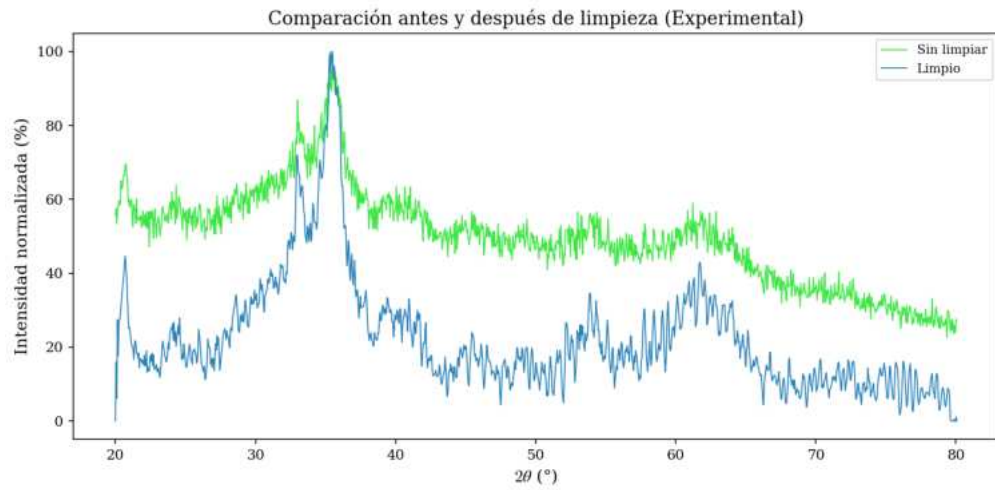


Figure 18: *Theoretical XRD pattern for $\alpha\text{-Fe}_2\text{O}_3$ (hematite phase, COD-2101167). Trigonal rhombohedral structure with $a = b = c = 5.43 \text{ \AA}$, $\alpha = \beta = \gamma = 55.28^\circ$. The most intense peak appears near $2\theta = 33.16^\circ$.*

Data were processed using PreDICT software, applying a 9-point smoothing window and removing the $K\alpha_2$ contribution.



(a) High-resolution pattern before and after processing.



(b) Experimental pattern before and after processing.

Figure 19: Diffractograms before and after data cleaning. Processing reduces noise and background without shifting peak positions.

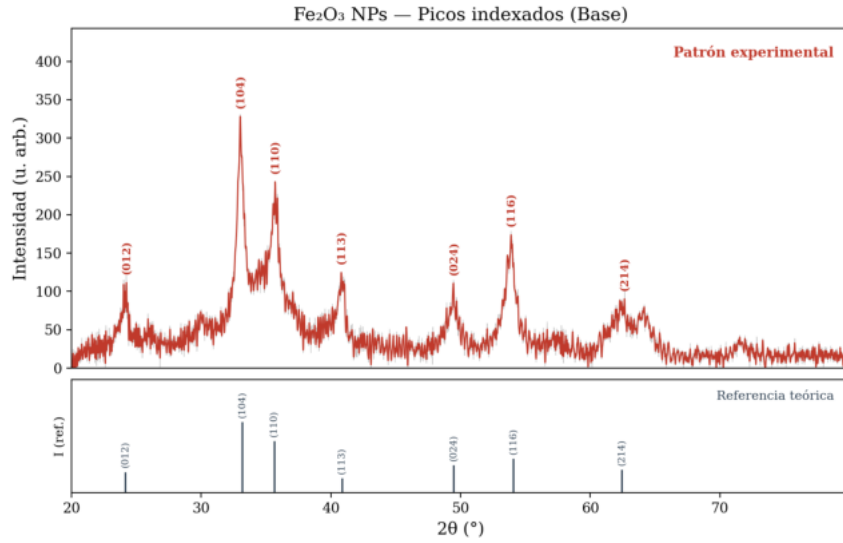


Figure 20: XRD pattern of Fe_2O_3 nanoparticles with indexed peaks — high-resolution dataset.

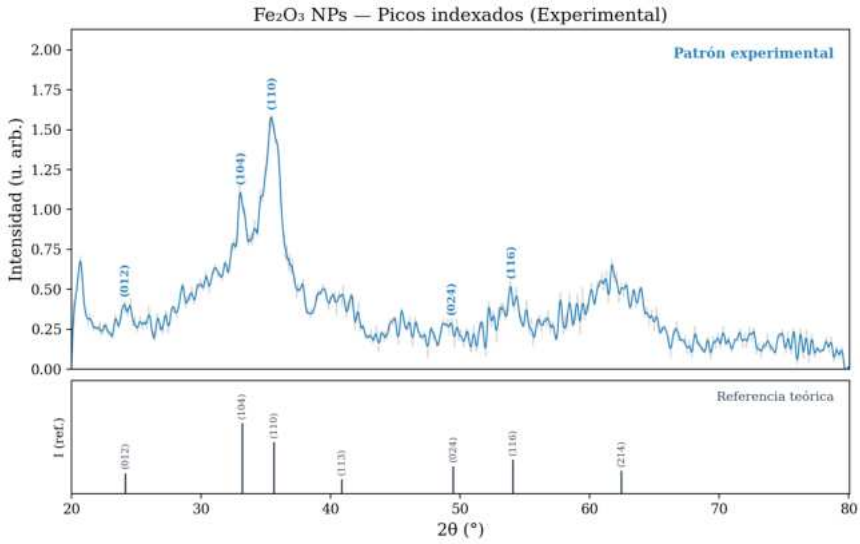


Figure 21: XRD pattern of Fe_2O_3 nanoparticles with indexed peaks — experimental dataset.

Peak broadening relative to the theoretical pattern is evident in both datasets, consistent with nanometric crystallite sizes. The experimental pattern shows fewer resolved peaks and a broader amorphous hump, characteristic of very small crystallites. Relative peak intensities differ from the theoretical pattern, indicating some preferential orientation and size distribution effects.

Texture coefficient (Harris method)

Table 3: Texture coefficients for the high-resolution pattern.

(hkl)	$2\theta_{\text{exp}} (^{\circ})$	$2\theta_{\text{ref}} (^{\circ})$	I_{exp}	$I_{0,\text{ref}}$	I/I_0	$TC(hkl)$
(012)	24.23	24.15	111.48	29.23	3.8139	1.0400
(104)	33.01	33.16	328.18	100.00	3.2818	0.8949
(110)	35.70	35.63	242.92	72.95	3.3298	0.9080
(113)	40.80	40.86	124.72	20.67	6.0338	1.6454
(024)	49.46	49.46	110.65	38.85	2.8483	0.7767
(116)	53.89	54.07	173.77	48.33	3.5956	0.9805
(214)	62.64	62.44	90.30	32.64	2.7664	0.7544

Table 4: Texture coefficients for the experimental pattern.

(hkl)	$2\theta_{\text{exp}} (^{\circ})$	$2\theta_{\text{ref}} (^{\circ})$	I_{exp}	$I_{0,\text{ref}}$	I/I_0	$TC(hkl)$
(012)	24.08	24.15	0.41	29.23	0.0139	1.0726
(104)	33.05	33.16	1.11	100.00	0.0111	0.8564
(110)	35.42	35.63	1.58	72.95	0.0216	1.6717
(024)	49.32	49.46	0.29	38.85	0.0074	0.5734
(116)	53.91	54.07	0.52	48.33	0.0107	0.8260

The high-resolution pattern shows slight preferential orientation in the (113) plane ($TC = 1.6454$); the experimental pattern shows a more pronounced preference in the (110) plane ($TC = 1.6717$). Neither sample exhibits strongly textured behavior.

Crystallite size — Scherrer method

FWHM values were extracted from smoothed diffractograms (Savitzky–Golay filter) using linear interpolation of the half-maximum intersections, with baseline correction applied to each peak.

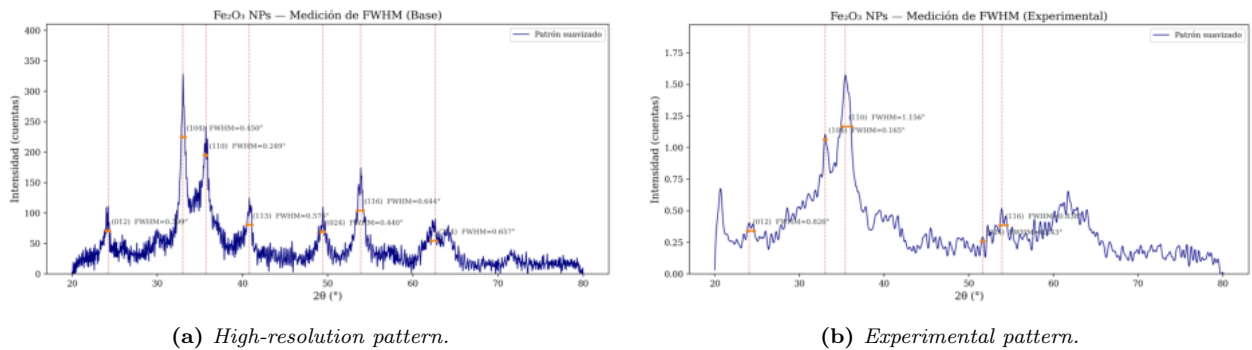


Figure 22: FWHM estimation for crystallite size calculation via the Scherrer equation.

Table 5: *Crystallite sizes from the Scherrer equation — high-resolution pattern.*

(hkl)	2θ (°)	FWHM (°)	β_{corr} (rad)	D (nm)
(012)	24.236	0.3989	0.006962	20.14
(104)	33.006	0.4502	0.007858	18.20
(110)	35.700	0.2488	0.004342	33.17
(113)	40.797	0.5761	0.010056	14.55
(024)	49.455	0.4397	0.007674	19.67
(116)	53.886	0.6439	0.011238	13.69
(214)	62.635	0.6575	0.011475	13.99

The mean crystallite size for the high-resolution pattern is $D_{\text{avg}} = 19.06 \pm 6.79$ nm.

Table 6: *Crystallite sizes from the Scherrer equation — experimental pattern.*

(hkl)	2θ (°)	FWHM (°)	β_{corr} (rad)	D (nm)
(012)	24.079	0.8256	0.014409	9.73
(104)	33.049	0.1652	0.002883	49.61
(110)	35.417	1.1556	0.020168	7.14
(024)	51.643	0.2434	0.004248	35.86
(116)	53.910	0.8379	0.014624	10.52

The mean crystallite size for the experimental pattern is $D_{\text{avg}} = 22.5 \pm 19.08$ nm. Both values confirm the nanometric character of the catalyst particles, which is favorable for CNT growth: smaller catalyst particles promote the formation of finer, better-distributed nanostructures.

Williamson–Hall analysis

Table 7: *Crystallite size and microstrain from the Williamson–Hall method.*

Sample	D_{WH} (nm)	ε	R^2
High-resolution	39.57	3.00×10^{-3}	0.3691
Experimental	8.55	4.04×10^{-3}	0.0569

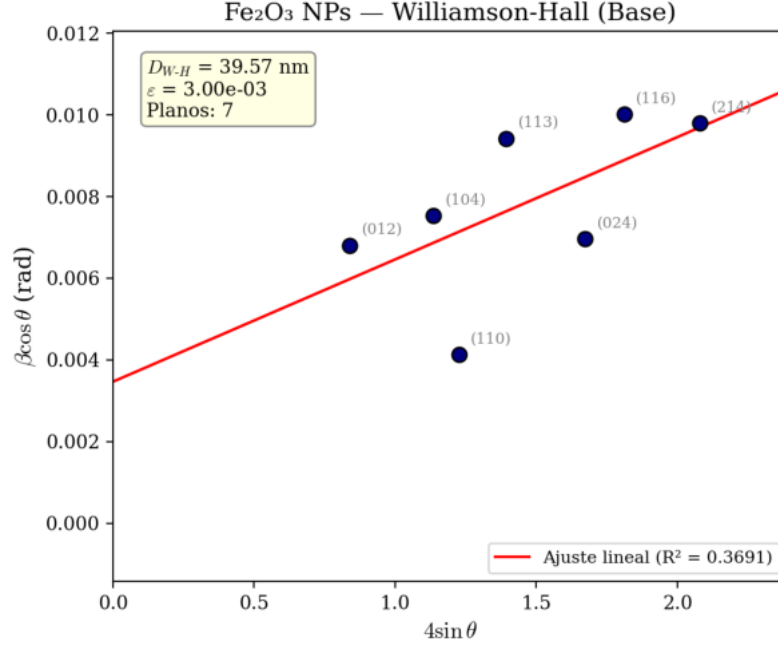


Figure 23: *Williamson-Hall plot for the high-resolution pattern.*

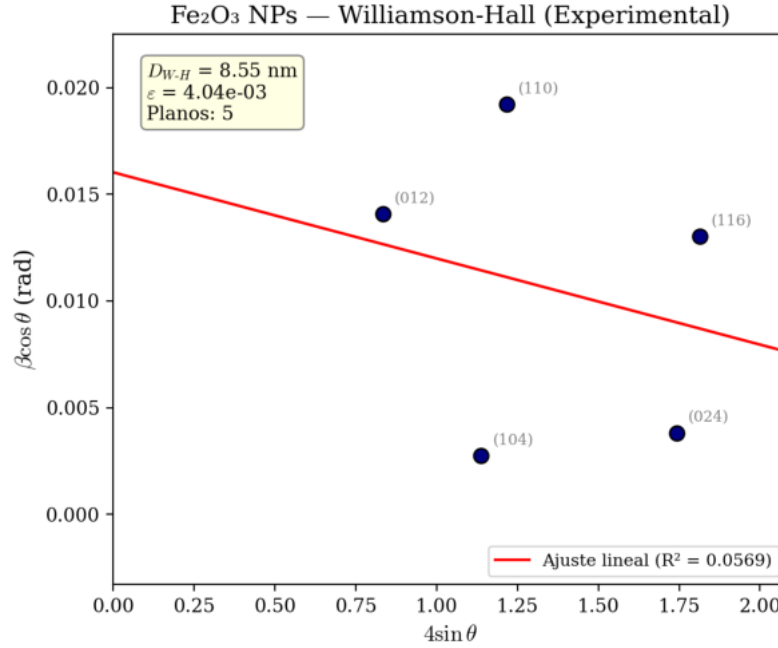


Figure 24: *Williamson-Hall plot for the experimental pattern.*

Crystallite sizes from the Williamson-Hall method are in the nanometric range for both datasets. The microstrains are small and effectively negligible. The low R^2 values (< 0.37) indicate poor linear correlation, so the Williamson-Hall model does not represent the crystallite size distribution reliably in this case — a result likely caused by the broad size dispersion inherent to the nanoparticle powder.

FTIR Spectroscopy

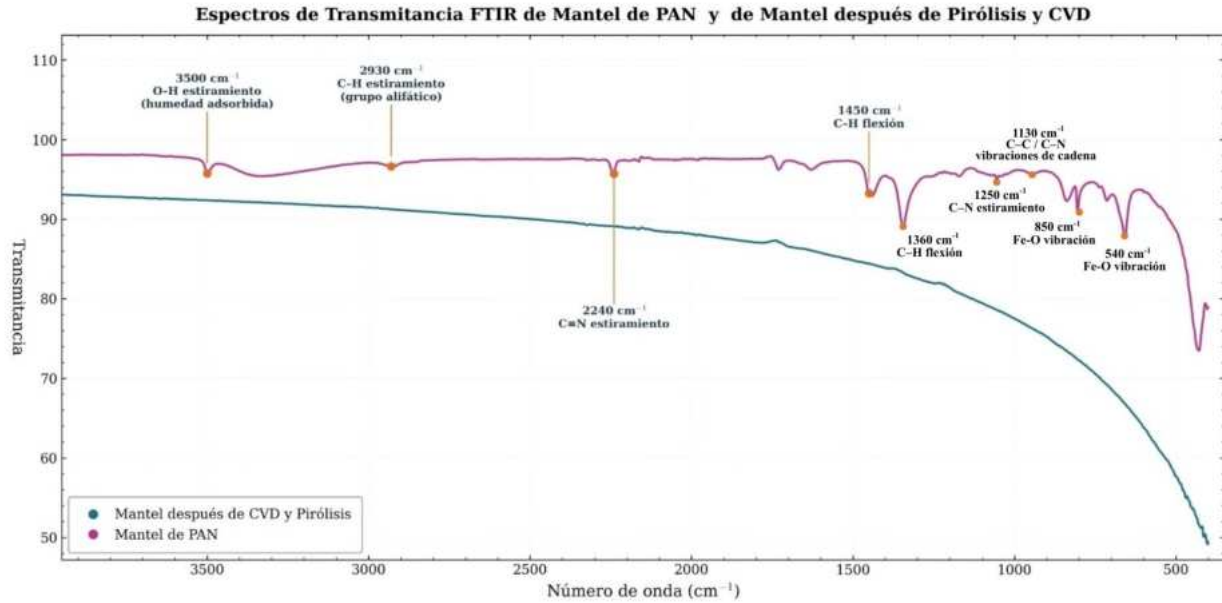


Figure 25: FTIR spectra: PAN precursor (top) and carbonized mat (bottom).

The FTIR spectra allow tracking of the chemical transformation from PAN to carbonized material. The precursor spectrum displays several characteristic bands:

Table 8: Characteristic FTIR bands identified in the PAN precursor spectrum.

Wavenumber (cm^{-1})	Vibrational assignment	Description
~ 3500	O-H stretching	Adsorbed moisture
~ 2930	C-H stretching	Aliphatic groups
~ 2240	$\text{C}\equiv\text{N}$ stretching	Nitrile group — characteristic of PAN
~ 1450	C-H bending	Structural deformation
~ 1360	C-H bending	Structural deformation
~ 1250	C-N stretching	Carbon-nitrogen bond in PAN
~ 1130	C-C / C-N vibrations	Polymer backbone
~ 850	Fe-O vibration	Iron-oxygen bond in iron oxides
~ 540	Fe-O vibration	Characteristic of the Fe_2O_3 lattice

The most diagnostic band is at $\sim 2240 \text{ cm}^{-1}$, corresponding to $\text{C}\equiv\text{N}$ stretching — the nitrile group that uniquely identifies PAN. The broad O-H band at $\sim 3500 \text{ cm}^{-1}$ is attributed to environmental moisture adsorbed during sample handling, not to the material itself.

The carbonized mat spectrum (blue) shows near-complete disappearance of the PAN characteristic bands, particularly the nitrile group. This drastic change confirms that CVD and pyrolysis successfully transformed the polymer into a carbon-dominated structure, through elimination of organic functional groups and reorganization of the molecular backbone into a graphitic network. The FTIR results are fully consistent with the EDS elemental analysis.

Raman Spectroscopy

The raw spectrum was trimmed to 100–3000 cm^{-1} (removing a cosmic ray spike near 3200 cm^{-1}) and smoothed using a Savitzky–Golay filter (window: 11 points, polynomial order: 3).

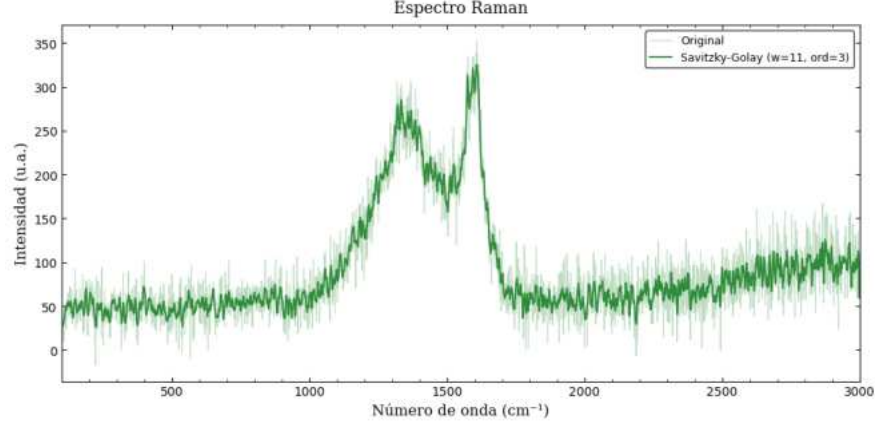


Figure 26: Raman spectrum of the carbonized fiber mat after trimming and smoothing.

The spectrum displays two dominant features: the G band near 1580–1600 cm^{-1} , associated with in-plane C–C stretching in ordered graphitic structures, and the D band near 1350 cm^{-1} , associated with structural defects and disorder. The G band intensity is slightly higher than the D band, indicating a partially graphitized structure with a significant defect density. The relatively broad peak widths suggest small graphitic domains.

The intensity ratio

$$\frac{I_D}{I_G} = \frac{284.85}{325.17} = 0.8760 \quad (8)$$

allows estimation of the graphitic domain size via the Tuinstra–Koenig relation,

$$L_a \text{ (nm)} = C(\lambda) \left(\frac{I_D}{I_G} \right)^{-1}, \quad (9)$$

where $C(\lambda) = 4.4 \text{ nm}$ for a 514 nm excitation wavelength, giving

$$L_a = 5.02 \text{ nm}. \quad (10)$$

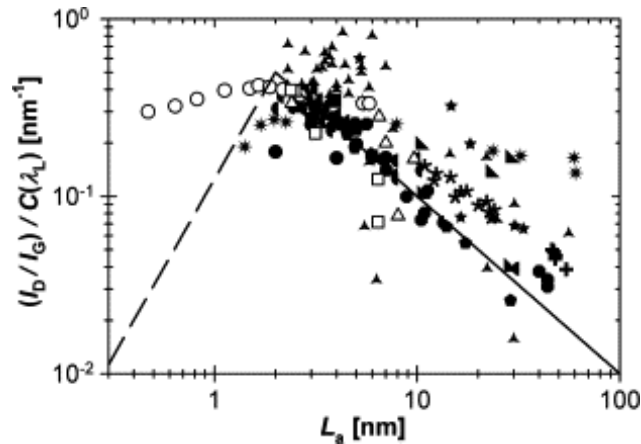


Figure 27: Correlation between L_a from XRD and the normalized Raman ratio $(I_D/I_G)/C(\lambda)$ for various carbon materials. PAN-based carbon fibers are indicated by ($\square$). Adapted from Zickler et al. (2006).

The estimated $L_a = 5.02$ nm is consistent with PAN-based carbon fiber literature values (Figure 27). The absence of the G' (2D) band near 2700 cm^{-1} — characteristic of highly ordered structures such as graphene and single-walled CNTs — is coherent with this small domain size: graphitic domains of ~ 5 nm generate a 2D signal too weak and broad to distinguish from the noise baseline.

Importantly, the absence of the 2D band does not imply the absence of CNTs. The high structural disorder of the carbon fiber matrix masks the Raman signal that CNTs would otherwise emit. Direct morphological characterization by SEM is therefore necessary to confirm their presence.

Complementarity of FTIR and Raman spectroscopy. FTIR and Raman are complementary vibrational spectroscopies. FTIR is most sensitive to vibrations involving dipole moment changes, making it ideal for identifying functional groups and surface chemistry. Raman is most sensitive to vibrations involving changes in electronic polarizability, making it particularly suited for analyzing the sp^2 carbon backbone — specifically the D and G bands that report on defect density and graphitization degree. Used together, the two techniques provide a comprehensive chemical and structural picture of the carbonized material.

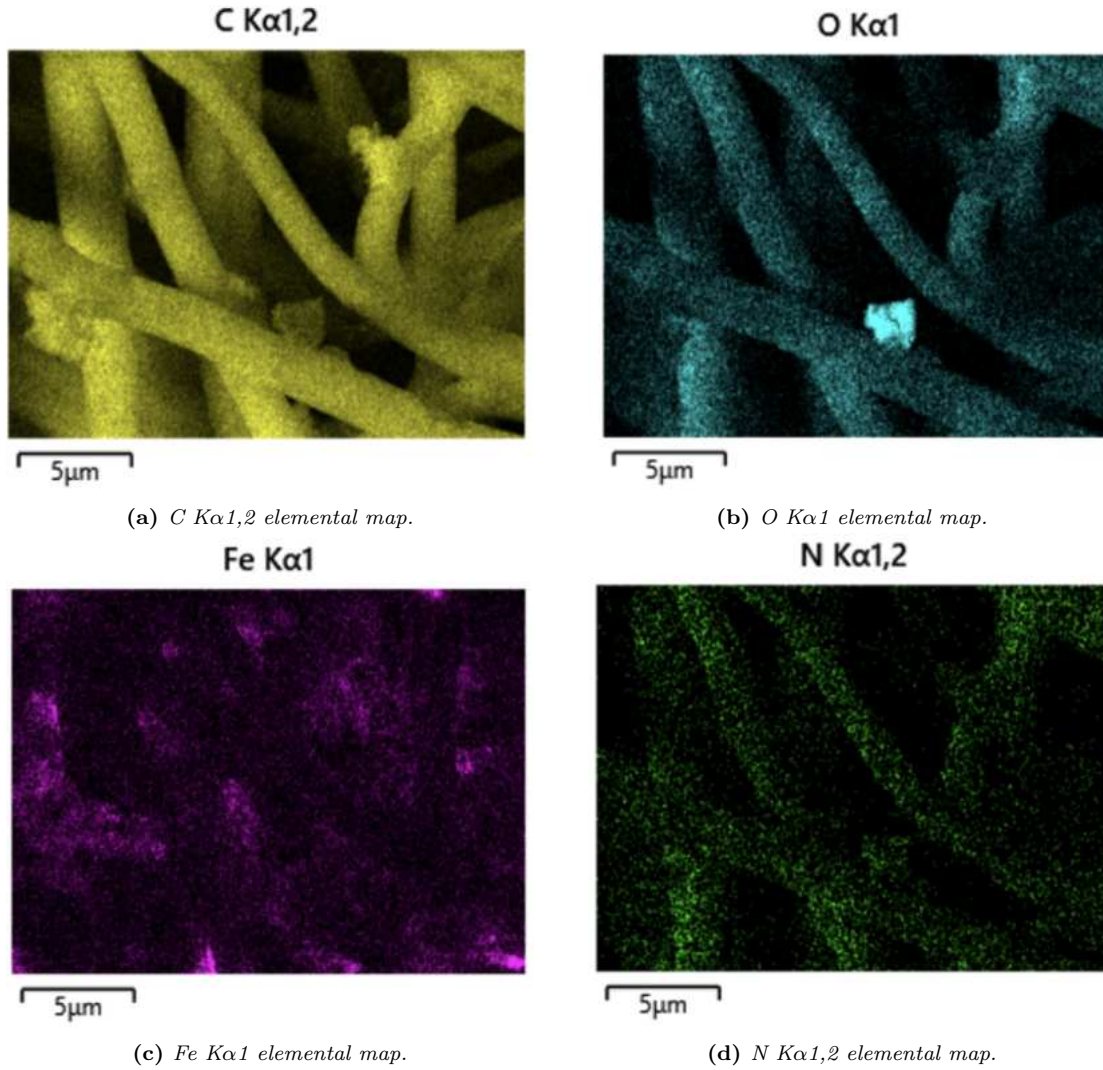


Figure 31: EDS elemental maps of the carbonized fiber mat showing the distribution of C, O, Fe, and N.

Table 9: Elemental composition of the sample obtained by EDS.

Element	Composition (wt%)
Carbon (C)	80.0
Oxygen (O)	9.1
Iron (Fe)	8.0
Nitrogen (N)	2.8

EDS elemental analysis. The composition is dominated by carbon (80 wt%), consistent with successful carbonization of the PAN precursor and camphor-derived carbon from CVD. Oxygen (9.1 wt%) is attributed to surface oxidation and to the iron oxide nanoparticles. Iron (8.0 wt%) confirms the presence of the catalyst particles in the final material. Nitrogen (2.8 wt%) originates from PAN, as part of the polymer's nitrogen content is retained in the carbon lattice after thermal treatment.

Conclusions

The comprehensive characterization of carbon fiber mats with surface-grown CNTs confirms the successful formation of the target material, while also identifying areas where the synthesis process can be improved to enhance structural homogeneity.

FTIR spectroscopy provided clear evidence of the chemical transformation induced by CVD and pyrolysis: the near-complete disappearance of the PAN characteristic bands — most notably the $\text{C}\equiv\text{N}$ nitrile peak at $\sim 2240\text{ cm}^{-1}$ — confirms that the thermal treatment achieved its intended effect. This result is further corroborated by EDS, which shows a carbon-dominated composition.

Raman spectroscopy revealed partially graphitized carbon with small graphitic domains ($L_a \approx 5.02\text{ nm}$) and a significant defect density, consistent with literature values for PAN-derived carbon fibers. The absence of the 2D band reflects the structural disorder of the fiber matrix rather than the absence of CNTs: the disorder masks the Raman signal that CNTs would otherwise produce.

SEM micrographs confirmed the presence of nanometric filamentary structures with diameters of 116–165 nm on the fiber surfaces, consistent with CNT growth catalyzed by the iron nanoparticles. EDS corroborated the catalytic role of the Fe nanoparticles, though the non-uniform spatial distribution of CNT growth suggests that process optimization — particularly of the nanoparticle concentration and the camphor sublimation and temperature ramp profiles — is needed.

XRD and UV–Vis established that the Fe_2O_3 catalyst nanoparticles present nanometric crystallite sizes ($\sim 19\text{ nm}$ by Scherrer) and a band gap of 2.16–2.23 eV, consistent with the hematite phase.

Overall, the results validate the proposed synthesis route — electrospinning followed by pyrolysis and CVD with Fe_2O_3 catalyst — as a viable method for producing carbon fiber mats with CNTs. The observed structural inhomogeneity motivates further systematic investigation of the synthesis parameters to improve the uniformity and quality of CNT growth.

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